The RESOLVE Swiss trial for people with chronic low back pain: statistical analysis plan for a cluster randomised controlled trial

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Abstract

Background. Statistical analysis plans describe the planned data management and analysis of a clinical trial before the data are seen, and so protect the eventual results from data-driven analytical choices. This paper reports the statistical analysis plan for the RESOLVE Swiss trial, which evaluates graded sensorimotor retraining combined with pain neuroscience education against usual physiotherapy for people with chronic low back pain.

Design and analysis. RESOLVE Swiss is a two-arm parallel-group *cluster* randomised controlled trial in Swiss physiotherapy practices. The practice is the unit of randomisation and the patient is the unit of analysis. The primary outcome is back-specific disability, measured with the Roland–Morris Disability Questionnaire at 18 weeks after randomisation. The primary analysis is a linear mixed model over the baseline and 18-week measurements with fixed effects for time and the treatment-by-time interaction, no main effect of treatment, and random intercepts for practice and participant. The arms share a common baseline mean by design, which is what lets the model use the baseline information, and the treatment effect is estimated with Kenward–Roger degrees of freedom. A supplementary Bayesian analysis with prespecified priors, informed by the individual participant data of the Australian trial, accompanies the frequentist results. Because the trial randomises a small number of practices, and because

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mixed-model inference is anti-conservative in that setting, we prespecify two further analyses of the primary outcome: a cluster-level (two-stage) analysis and an exact randomisation test over the finite set of possible allocations. We report the intra-cluster correlation with its uncertainty, the analyses of secondary outcomes, the handling of missing data, non-adherence and contamination, and the planned reporting.

Conclusion. Prespecifying the analysis reduces the scope for bias in the results of RESOLVE Swiss and makes explicit how the small number of clusters is dealt with.

Trial registration. [OPEN: registry and registration number]

Keywords: Low back pain; Cluster randomised trial; Statistical analysis plan; Mixed models; Physical therapy.

1 Introduction

1.1 Background and rationale

Chronic low back pain is among the largest contributors to years lived with disability worldwide, and the effects of the treatments physiotherapists currently offer are, on average, modest. The Australian RESOLVE trial tested whether a treatment aimed at the central nervous system representation of the back (graded sensorimotor precision training preceded by pain neuroscience education) reduces pain more than a sham intervention, and reported a small but consistent benefit at 18 weeks [1]. An earlier and much smaller Swiss pilot compared a related programme with usual physiotherapy and pointed in the same direction without being able to resolve the question [2].

Two things are unresolved. First, RESOLVE compared the intervention with a sham. The clinically relevant comparison for Swiss practice is against usual physiotherapy as it is actually delivered. Second, the intervention has so far been delivered in a trial setting by trial therapists. RESOLVE Swiss therefore delivers it in ordinary physiotherapy practices, by the therapists who work there. That choice makes the trial pragmatic, and it forces the design to be a *cluster* randomised trial: once a practice has been trained in the intervention, its therapists cannot un-learn it for the next patient, so allocation has to happen at the level of the practice rather than the patient [3].

Cluster randomisation changes the statistics in ways that are easy to get wrong, and the difficulty is sharpened here because the number of practices is small. This analysis plan is written before the analysis of outcome data and follows the recommendations for the content of statistical analysis plans in clinical trials [4], together with the CONSORT extension for cluster randomised trials [3]. It reports the planned frequentist analysis of the primary and secondary outcomes, a supplementary Bayesian analysis of the primary outcome, and a summary of the Bayesian interim analysis that the protocol specifies.

2 Methods

2.1 Trial design

RESOLVE Swiss is a two-arm, parallel-group, cluster randomised, investigator blinded, multicentre trial with 1:1 intended allocation at the level of the physiotherapy practice [5]. Practices and institutions are the clusters. Patients treated within a practice are the units of observation and of analysis. Physical outcomes are assessed at baseline and at 18 weeks. All self-reported questionnaires, including the primary outcome, are collected again at 26 and 52 weeks after the baseline measurement.

The protocol plans 20 physiotherapy institutions across Switzerland. At the time of writing [OPEN: confirm] 5 practices are allocated to the intervention arm and 9 to the control arm, a deviation from the planned number and from the 1:1 ratio. The consequences of that split, and of any change to it, are quantified in section 2.6. The deviation has to be reported under the CONSORT cluster extension [3] whether or not it is corrected.

The trial is a Category A study under the Swiss Ordinance on Clinical Trials (ClinO Art. 61). Ethical approval was granted by [OPEN: ethics committee and approval number]. The trial will be registered at ClinicalTrials.gov and in the Swiss National Clinical Trial Portal [OPEN: registration numbers once issued].

2.2 Eligibility

A cluster randomised trial has two eligibility layers, and both are reported here because selection can operate at either [3].

Practices. Swiss outpatient physiotherapy practices and institutions willing to be allocated to either arm. Therapists in practices allocated to the intervention must complete a five-day RESOLVE training course and have their skills assessed before delivering any study treatment [OPEN: confirm any further practice-level criteria, e.g. minimum patient volume].

Participants. Adults aged 18 to 75 referred to physiotherapy for low back pain, with non-specific low back pain persisting for at least three months, a baseline RMDQ of at least 7 points, sufficient German to follow the study procedures and complete the questionnaires, internet access and willingness to use a private device, and a person available to assist with home training. Participants are excluded for evidence of specific spinal pathology (malignancy, fracture, infection, inflammatory joint or bone disease), radiculopathy, pregnancy or being less than six months postpartum, a coexisting major medical disease contraindicating general exercise, spinal surgery within the preceding 12 months, an intra-articular or perineural lumbar steroid injection within the preceding three months, inability to follow the study procedures, or an open legal dispute with an insurer relating to back pain.

An fMRI substudy enrols 40 of the 220 participants. Its analysis is exploratory and is not covered by this plan.

2.3 Interventions

Participants in both arms receive 12 one-to-one sessions with a qualified physiotherapist over 16 weeks. The first two sessions may last an hour and the remaining ten about 30 minutes, weekly at first and then fortnightly. The intervention is the RESOLVE programme (pain science education, graded sensory retraining, and graded motor relearning and loading), delivered by therapists who have completed the five-day training. The comparator is usual physiotherapy, delivered by therapists who have not.

Usual physiotherapy is expected to be heterogeneous between practices. What control practices actually deliver will be recorded and described, because in a pragmatic trial the comparator is part of the finding rather than a nuisance.

2.4 Randomisation and allocation

Practices are allocated 1:1 by the central trial coordinator at ZHAW, before the intervention therapists are trained. The allocation sequence is concealed from everyone involved in recruitment and enrolment, and the randomisation table is password protected. The protocol specifies neither stratification nor covariate-constrained allocation [OPEN: confirm that allocation was simple].

Two points follow for the analysis. First, if any variable was in fact used to stratify or constrain the allocation, it must be adjusted for in the analysis, otherwise the standard errors are wrong [6, 7]. If allocation was simple, no such adjustment is required and the randomisation test of section 3.2.3 may enumerate all allocations. Second, if practices were recruited in waves and later waves were allocated non-randomly, wave and treatment arm would be confounded. Recruitment wave will therefore be recorded for every practice and, if practices entered in distinct waves, carried in the model as a covariate.

Participants cannot be blinded, since the intervention includes a substantial home training component. The three physical assessments (two-point discrimination, the point-to-point test and lumbar movement control) are video-recorded and rated by a blinded assessor. The remaining outcomes are self-reported and completed at home through REDCap, which removes interviewer influence. [OPEN: confirm that the statistician will be blinded to arm labels until the primary analysis is complete]

2.5 Recruitment of participants after cluster allocation

Because practices are randomised before their patients are recruited, and because the recruiting therapists know which arm their practice is in, patient selection can differ between arms. This identification or recruitment bias is the characteristic threat to cluster randomised trials and is not present in individually randomised trials [3, 8]. We will therefore report, by arm and by practice, the number of patients screened, found eligible, approached, and consenting, and we will compare baseline characteristics between arms with this threat explicitly in mind. Baseline imbalance in a cluster randomised trial with few clusters is expected by chance and is not by itself evidence of bias. It is, however, the main signal that recruitment differed between arms, and it will be interpreted alongside the screening counts rather than by significance testing [3, 9].

2.6 Sample size

The target difference is 2 points on the Roland–Morris Disability Questionnaire (RMDQ, range 0–24), the smallest between-group difference the investigators consider clinically worthwhile in this population [5]. The calculation below is a rederivation of the protocol's sample size under the hypothesis framing set out in section 2.8. It lands on essentially the number the protocol already commits to, by a route that a reader can reproduce. The between participant standard deviation of the RMDQ at 18 weeks is taken from the Australian trial (intervention 3.6, SD 4.6; control 6.4, SD 5.8 [1]), giving a pooled SD of 5.2 points. With a two-sided $\alpha = 0.05$ and 80% power, an individually randomised trial would need 108 participants per arm. Using the baseline measurement, with an assumed baseline-to-follow-up correlation of 0.6, reduces this to 69 per arm.

Clustering inflates this requirement by the design effect. Allowing for unequal practice sizes [10],

$$\mathrm{DE} = 1 + \left[\left(\mathrm{cv}^2\,\frac{k-1}{k} + 1\right)\bar{m} - 1\right]\rho, \tag{1}$$

where $\bar{m}$ is the mean number of analysed participants per practice, k the number of practices, ρ the intra-cluster correlation and cv the coefficient of variation of practice size. Intra-cluster correlations reported for primary care outcomes are usually small, with a median near 0.01 [11]. Unequal cluster size can be ignored when $\mathrm{cv} < 0.23$, but for trials randomising general practices cv is typically around 0.65 [10]. Physiotherapy practices are the closest analogue we have, so we plan on $\mathrm{cv} = 0.65$ and report the more optimistic 0.4 as well. With $\bar{m} = 14$, $k = 14$ and $\mathrm{cv} = 0.65$, the design effect is 1.19 at $\rho = 0.01$, 1.56 at $\rho = 0.03$ and 1.93 at $\rho = 0.05$, so the requirement is 82, 107 and 133 participants per arm respectively. The trial therefore aims for 100 analysed participants per arm, which corresponds to 110 recruited per arm allowing 10% loss to follow-up (Table 1). This covers an intra-cluster correlation up to about 0.03.

Two approximations in this calculation should be named. The variance reduction from the baseline measurement is applied as the individual-level factor $1 - r^2$ and the clustering inflation as (1). A formula that handles baseline adjustment inside a cluster randomised design, using the individual-level and the cluster-level baseline correlations separately, is available and gives a slightly smaller requirement when the cluster-level correlation is high [12]. Our version is therefore mildly conservative. Second, the intra-cluster correlation is assumed, not known.

The calculation is reported across a range of values rather than at a single one, which is what sensitivity to this assumption requires [7].

Three features of the current design should be stated plainly rather than buried. First, with 14 practices the between-cluster component of the variance rests on 14 numbers, and the relevant reference distribution has roughly 12 degrees of freedom. Power is correspondingly lower than the design effect alone suggests. Holding the analysed sample at 200 participants and using a t reference distribution on $k_1 + k_2 - 2$ degrees of freedom, the power to detect a 2-point difference is 77% at $\rho = 0.01$, 65% at $\rho = 0.03$ and 56% at $\rho = 0.05$.

Second, rebalancing the existing practices from 5 versus 9 to 7 versus 7 recovers only about 3 percentage points of power at every intra-cluster correlation considered, because the total analysed sample is what it is. Equal allocation of clusters is the efficient choice when the intra-cluster correlation and the total variance are the same in both arms, and a deliberate imbalance is justified only when they are not [13]. We have no reason to expect them to differ here, so 7 versus 7 is preferable, but the efficiency argument for changing the split is weak.

Third, and this is the binding constraint: *the number of practices is too small.* Five clusters in one arm sits at the bottom of the range in which the recommended analyses have been evaluated [14], and a minimum of about ten clusters per arm has been recommended for cluster randomised trials generally, with the practical rule of adding two to three clusters per arm to any sample size derived from a normal approximation [15]. Few clusters also raise the probability of chance imbalance on practice characteristics and reduce the number of analyses available [16]. Adding practices while holding practice size at 14 participants raises power from 62% with 5 per arm to 80% with 7 per arm and 90% with 9 per arm at $\rho = 0.01$ (Table 2). That is a much better return than adding patients to the practices already enrolled, because increasing cluster size at a fixed number of clusters has strongly diminishing returns [17]. If additional practices are recruited, they must be randomised as a batch rather than assigned to the intervention arm, so that the new wave contains its own randomised contrast [18, 19].

All sample size and power computations are reproducible from the script `R/sample_size.R` that accompanies this paper.

2.7 Data integrity and missing data

Patient-reported outcomes are captured electronically in REDCap, completed by participants at home through a personalised link. The eCRF holds eligibility, demographic, medical-history and physical-assessment data. [OPEN: confirm the data-cleaning and query workflow, and who locks the database]

Data will be inspected for implausible values, range violations and duplicate records before the analysis begins, and the cleaned data set will be locked before any comparison between arms is made. We will report the amount and the pattern of missing outcome data by arm and by practice, since in a cluster trial an entire practice can be lost, which is a different problem from individual dropout.

2.8 Hypothesis framing: a clarification of the protocol

The protocol states the primary hypothesis as a superiority-by-a-margin test: the null hypothesis is that the intervention does *not* exceed usual physiotherapy by more than 2 RMDQ points, tested one-sided at $\alpha = 0.05$ [5]. This plan instead tests the conventional null of no difference, with 2 points as the *target* difference the trial is powered to detect. The change is deliberate and is recorded here rather than made silently.

The reason is that the two framings imply very different trials. Under the margin formulation the quantity that drives power is the amount by which the true effect exceeds 2 points, not the effect itself. The sample size of 110 per arm was obtained under that formulation with an assumed true effect of about 2.4 points and a standard deviation of about 0.9. That 0.9, however, is the precision-weighted spread of the pooled treatment-effect estimates from the two previous studies [1, 2]. It measures uncertainty about the true effect. The quantity the formula requires is the between-participant standard deviation of the RMDQ, which is about 5.2 points. Recomputed with 5.2, a margin-based design powered to detect an excess of 0.4 points over a 2-point null would require several thousand participants per arm. It is not attainable and was never what the trial was resourced for.

If the trial is instead read as a conventional superiority trial targeting a 2-point difference, 110 participants per arm is close to right (section 2.6). The protocol's intent is to detect a clinically meaningful difference of at least 2 points. We interpret that intent in the conventional way, and treat the margin wording as an infelicity rather than a design choice. [OPEN: this requires the sponsor's agreement and, if the wording in the registered protocol is to change, a protocol amendment; the alternative is to keep the margin formulation and state openly that the trial is underpowered for it]

The protocol also specifies a one-sided test for the primary outcome and two-sided tests for the secondary outcomes. A one-sided test at $\alpha = 0.05$ is equivalent to a two-sided test at 0.10 in the direction of interest and is generally discouraged in trial reports. We prespecify two-sided tests throughout and report the estimate with its two-sided 95% confidence interval, which is the more conservative choice and does not affect the direction of any conclusion. [OPEN: sponsor to confirm]

2.9 Analytic principles

- Participants will be analysed in the arm to which their practice was allocated, regardless of the treatment actually delivered (intention to treat).
- All treatment effects will be reported as point estimates with 95% confidence intervals. Two-sided tests at a nominal α of 0.05 accompany the primary and secondary analyses, but the interpretation will rest on the estimate and its interval rather than on whether a threshold was crossed [20, 21].
- Confidence intervals and p values will not be adjusted for multiplicity. There is one primary outcome at one time point. Secondary outcomes are explicitly labelled as secondary and will be interpreted as such [4].
- Every analysis accounts for clustering by practice. No analysis will treat patients from the same practice as independent.
- Analyses will be carried out in R [22], with mixed models fitted using `lme4` [23] and small-sample inference obtained with `lmerTest` and `pbkrtest` [24, 25]. Analysis code will be published with the results.

2.10 Outcome definitions

2.10.1 Primary outcome

Back-specific disability at 18 weeks after randomisation, measured with the Roland–Morris Disability Questionnaire, a 24-item instrument scored from 0 (no disability) to 24 [26, 27].

2.10.2 Secondary outcomes

The protocol specifies a large number of secondary endpoints, which we group here by the analysis they receive.

Patient-reported, continuous, at 18, 26 and 52 weeks. Pain intensity (numeric rating scale); Patient Specific Functional Scale; Work Productivity and Activity Impairment; Neurophysiology of Pain Questionnaire; Back Pain Attitudes Questionnaire; Pain Self-Efficacy Questionnaire; Depression Anxiety and Stress Scale; Tampa Scale of Kinesiophobia; EQ-5D-5L; Insomnia Severity Index; Fremantle Back Awareness Questionnaire; Goal Attainment Scale; the epistemic-beliefs questionnaire; and the treatment expectation, perceived improvement and satisfaction items.

Physical, continuous, at 18 weeks only. Two-point discrimination and the point-to-point test of tactile acuity, and lumbar movement control, each rated from video by a blinded assessor.

Counts and other. Medication intake, treatment adherence, and adverse events.

Cost-effectiveness (quality-adjusted life years and incremental cost-effectiveness ratios) and the fMRI substudy are reported separately and are not covered by this plan. [OPEN: confirm that the health-economic analysis is out of scope here]

3 Analysis

3.1 Baseline description

We will describe the sample at two levels, as the CONSORT cluster extension requires [3]. At the practice level (Table 3, panel A) we will report the number of practices per arm, the number of therapists, practice size, urban or rural setting, and the number of participants contributed. At the participant level (panel B) we will report means and standard deviations for approximately normally distributed continuous variables, medians and interquartile ranges otherwise, and counts with percentages for categorical variables. Baseline characteristics will not be tested for statistical significance between arms: the null hypothesis of no difference is true by construction under randomisation, so such a test answers no question anyone is asking [9, 28].

3.2 Primary outcome

3.2.1 Estimand

The quantity of interest is the *participant-average* difference in mean RMDQ at 18 weeks between the two arms: the difference that would be seen if every patient of this kind in these practices were treated one way rather than the other, with each patient counting equally. In a cluster randomised trial with unequally sized clusters this is not the same quantity as the cluster-average effect, in which each practice counts equally, and different analyses target different quantities [29]. The choice is made here, at the design stage, as the consensus extension of the ICH E9(R1) estimand addendum to cluster randomised trials requires [30]. We prespecify the participant-average effect and report the cluster-average effect alongside it.

One consequence deserves stating because it constrains how the primary model below may be read. A mixed model with a random intercept weights clusters by their inverse variance, which is neither participant-average nor cluster-average weighting. When cluster size is informative (that is, when outcomes or treatment effects vary systematically with practice size), such a model targets neither estimand cleanly [29]. The cluster-level analyses specified in section 3.2.3 recover each estimand explicitly through their weighting, which is a second reason to report them rather than treating them as a mere robustness check. We will report the distribution of practice sizes so that readers can judge how much this matters here.

3.2.2 Primary model

Let y_{ijt} denote the RMDQ score of participant i in practice j at occasion t, where t indexes the baseline and 18-week measurements, let s_t equal 0 at baseline and 1 at 18 weeks, and let $x_j = 1$ if practice j was allocated to the intervention and 0 otherwise. The primary analysis is the linear mixed model

$$y_{ijt} = \beta_0 + \beta_1 s_t + \beta_2\, s_t x_j + u_j + w_{ij} + \varepsilon_{ijt}, \tag{2}$$

with a random intercept $u_j \sim N(0, \tau^2)$ for practice, a random intercept $w_{ij} \sim N(0, \nu^2)$ for participant within practice, and residual $\varepsilon_{ijt} \sim N(0, \sigma^2)$. The treatment effect at 18 weeks is β_2. The model deliberately contains no main effect of treatment. Keeping an interaction while dropping its main effect usually violates the variable inclusion principle, under which the interaction coefficient stops measuring what it should, because the omission imposes a constraint that is in general false [31]. Here the omitted term is the between-arm difference at baseline. The baseline is measured before allocation, so that difference is zero by design, the constraint is true rather than assumed, and β_2 still estimates exactly the 18-week treatment effect. Imposing the constraint is what lets the model use the baseline information. This constrained longitudinal formulation [32] matches the precision of adjusting the 18-week score for its baseline value, and one of the two is the recommended way to use a pre-randomisation measurement of the outcome in a cluster randomised trial [33]. The script `R/equivalence_constrained_vs_ancova.R` demonstrates the equivalence empirically. Both formulations beat analysing change scores, which are less precise whenever baseline and follow-up are positively correlated and inherit regression to the mean [34, 35]. With few clusters the main-effects form used here is also preferable to models carrying treatment-by-covariate interactions, which spend degrees of freedom and require a robust variance estimator [36].

Covariate adjustment in a cluster randomised trial is not automatically beneficial, unlike in an individually randomised one: each cluster-level covariate costs a degree of freedom, so that the degrees of freedom available for the treatment effect are approximately the number of practices minus two minus the number of cluster-level covariates [37]. With 14 practices the budget is 12. The baseline enters the model through the outcome vector and consumes no cluster-level degrees of freedom, and no further covariates are carried except the variables used to stratify or constrain the randomisation, which must be adjusted for or the type I error rate is not maintained [6, 7].

The model will be fitted by restricted maximum likelihood. Confidence intervals and p values for β_2 will use the Kenward–Roger approximation to the degrees of freedom and the associated small-sample adjustment to the covariance matrix of the fixed effects [38], which is the standard remedy for the anti-conservatism of naive mixed-model inference when the number of clusters is small [14, 39].

3.2.3 Inference with a small number of clusters

Uncorrected mixed models and generalised estimating equations have inflated type I error rates when the number of clusters is small, and the inflation is substantial: in a review and reanalysis of published cluster randomised trials, a large share of trials with few or moderately many clusters had used methods that do not maintain the nominal error rate [16, 39]. In a simulation across designs with 4 to 40 clusters, Leyrat and colleagues found that for continuous outcomes the Satterthwaite and Kenward–Roger corrections to a mixed model, and variance-weighted cluster-level analysis, did maintain the nominal rate, whereas uncorrected approaches did not [14]. Model (2) with Kenward–Roger degrees of freedom is therefore a defensible primary analysis for the RMDQ, and it is what the current guidance for cluster trials with fewer than about 40 clusters recommends [7, 40]. It is nonetheless an asymptotic argument resting on 14 numbers, so we prespecify two further analyses of the primary outcome, reported alongside the model-based result rather than instead of it. Cluster-level analyses, by contrast, need no such correction at all, which is why they are the natural companion when the number of clusters is very small [40].

Cluster-level analysis. We will regress y_{ij} on y_{ij}^0 within practices, take the practice-level mean residual as a single summary per practice, and compare the 14 summaries between arms with a two-sample t test on $k_1 + k_2 - 2 = 12$ degrees of freedom, weighting practices by their analysed size for the participant-average estimand and unweighted for the cluster-average estimand [29, 41]. This two-stage analysis gives up some efficiency but its type I error rate is correct by construction with any number of clusters.

Exact randomisation test. Because randomisation acts on practices, the set of allocations that could have occurred is finite and small: with 5 of 14 practices allocated to the intervention there are exactly $\binom{14}{5} = 2002$ of them, and with a 7 versus 7 split, 3432. We will therefore enumerate the full set, recompute the cluster-level test statistic under each allocation, and take the proportion of allocations giving a statistic at least as extreme as the observed one as an exact p value. This test makes no distributional assumption and is valid however few clusters there are [14, 42]. Its resolution is limited by the design: the smallest attainable two-sided p

value with a 5 versus 9 split is $2/2002 \approx 0.001$. If randomisation was covariate-constrained, the enumeration will be restricted to the allocations that satisfy the same constraints, since only those could have been selected [43].

If the three analyses agree, the model-based estimate is reported as the primary result. If they disagree, the disagreement itself will be reported and the randomisation test treated as the arbiter of the significance claim, since it is the only one of the three whose validity does not depend on the number of clusters.

3.2.4 Intra-cluster correlation

We will report the estimated intra-cluster correlation for the primary outcome, $\hat{\rho} = \hat{\tau}^2/(\hat{\tau}^2 + \hat{\sigma}^2)$, with a 95% confidence interval obtained by profile likelihood, and, if the interval is very wide, we will say so rather than presenting the point estimate alone. Reporting the observed intra-cluster correlation is a CONSORT requirement for cluster trials and is what makes the next trial in this field easier to design [3, 11].

3.3 The later time points

The primary model (2) extends naturally to all four occasions on which the RMDQ is collected (baseline, 18, 26 and 52 weeks), with time as a categorical variable, the same random intercepts, and again no main effect of treatment. We fit this four-occasion model to obtain the effects at 26 and 52 weeks, which are secondary. Its arm-by-time contrast at 18 weeks also serves as a check on the primary result.

Two differences from the protocol's model statement [5] are worth recording, and both are deliberate. The protocol carries a main effect of treatment alongside the interaction, which frees the baseline means of the two arms. We constrain them to be equal, which is true by design and recovers the precision that the sample size calculation assumes [32, 33]. And the protocol's statement does not mention a random effect for the practice. Omitting it would treat patients from the same practice as independent and understate the standard error, so a practice random effect is carried in every model in this plan. Mixed models for repeated measures behave acceptably in cluster randomised trials with continuous outcomes missing at random, including in simulations with as few as five clusters per arm [44].

3.4 Supplementary Bayesian analysis

The primary model (2) will also be fitted in a Bayesian formulation with the same structure. The prior for the treatment effect will be built from the individual participant data of the Australian trial, which the investigators hold under a data sharing agreement. The analogue of the primary model will be fitted to those data, and the resulting posterior for the treatment effect, downweighted to acknowledge that the Australian comparator was a sham rather than usual physiotherapy, will serve as the informative prior. The RMDQ at 18 weeks piles up near zero rather than spreading symmetrically, and access to the raw data allows the outcome distribution to be modelled directly instead of assumed normal. A sceptical prior centred at zero will be reported alongside the informative one.

With 14 practices the posterior for the between-practice variance is sensitive to its prior, which is the known weakness of Bayesian analysis of cluster randomised trials with few clusters [45]. We will therefore report a sensitivity analysis over the between-practice variance prior and, for the treatment effect, the posterior probabilities that the benefit exceeds 0 and 2 RMDQ points under each prior. The machinery matches the Bayesian interim analysis that the protocol specifies (section 3.9), so the interim and final analyses are consistent. The Bayesian analysis is supplementary. No claim of efficacy will rest on it alone, and the Australian data will not be redistributed with this plan.

3.5 Secondary outcomes

Secondary outcomes will be analysed with models of the same structure as (2), with the link function and distribution appropriate to the outcome, always including a random intercept for practice and always using the same small-sample correction. Effects will be reported as point estimates with 95% confidence intervals (Table 5). No formal claim of efficacy will be based on a secondary outcome.

3.6 Sensitivity analyses

The following are prespecified and will be reported whether or not they change the conclusion.

1. **Leave-one-practice-out.** The primary model refitted 14 times, each time omitting one practice. With few clusters a single atypical practice can carry the result, and the spread of

the 14 estimates shows how much it does.

2. **Therapist effects.** A third level for therapist within practice. This is the more faithful description of the data hierarchy, but variance components at three levels are poorly identified with 14 practices, so it is a sensitivity analysis rather than the primary model.
3. **Unconstrained model.** Model (2) with a main effect of treatment added, which frees the baseline means and shows what the constraint contributes.
4. **Cluster-mean baseline.** Model (2) with the practice mean baseline score added as a further fixed effect, which separates the within-practice from the between-practice association with baseline disability.
5. **Wave.** If practices entered in distinct recruitment waves, the primary model with wave as a fixed effect.

3.7 Missing data

The primary analysis uses all participants with a baseline measurement, whether or not the 18-week outcome was recorded, and is valid under a missing-at-random assumption conditional on the observed measurements. We will report the number and proportion of missing outcomes by arm and by practice, and compare the baseline characteristics of participants with and without an 18-week outcome. As a sensitivity analysis we will use multiple imputation by chained equations with the practice included in the imputation model, so that the imputation respects the clustering [44]. If an entire practice is lost, that will be reported separately and not imputed.

3.8 Non-adherence and contamination

Adherence will be summarised as the number of the 12 sessions attended and as the proportion of participants attending at least 75% of the intervention, which is the definition the protocol uses. Following the protocol, the mean effect in the principal stratum of participants who would adhere regardless of allocation, the complier average causal effect, will be estimated by instrumental variable regression with allocation as the instrument [46]. The estimate will be reported alongside the intention-to-treat effect, as in the Australian trial [1]. With 14 clusters this analysis is imprecise, and it is prespecified as exploratory.

Contamination is limited by design rather than only measured: the five-day RESOLVE training is unavailable to control therapists, and the German-language RESOLVE materials exist only inside a secured web application accessible to intervention therapists and participants. Treatment fidelity is monitored through documentation of every treatment session. We will nonetheless describe any contamination that comes to light and discuss its likely direction, since it biases the estimated effect towards the null [47].

3.9 Interim analysis

The protocol specifies a single Bayesian interim analysis for futility after 100 participants have been recruited [5]. The trial is declared futile if the posterior probability that the between-group difference is smaller than 1 RMDQ point reaches 95%, with the prior for the treatment effect built by precision-weighting the between-group differences of the previous trials. No sample size adjustment follows from the interim result. The supplementary Bayesian analysis of section 3.4 uses the same machinery and the same data sources, so the interim and final analyses are consistent.

4 Reporting

Results will be reported following CONSORT 2025 with the extension for cluster randomised trials [3, 48], including a flow diagram that tracks both practices and participants (Figure 1). Two items of the cluster extension have shaped this plan directly: items 7a and 17a require the intra-cluster correlation to be reported, with an indication of its uncertainty, at the design stage and for each primary outcome. Item 13a requires the flow diagram to give the number of clusters as well as the number of individuals at every stage [3].

The guideline for the content of statistical analysis plans that this document follows [4] contains no items specific to clustering. An extension for cluster randomised trials is under development but not yet published [49]. We have therefore added the cluster-specific items ourselves: the estimand and its weighting, the small-sample correction, the intra-cluster correlation, and the two-level description of the sample.

Declarations

Funding. [OPEN: funder and grant number]

Conflicts of interest. [OPEN: to be completed by all authors]

Ethics. [OPEN: committee, approval number, date]

Data and code availability. The code that reproduces the sample size and power calculations reported here accompanies this paper. Analysis code for the trial will be published together with the trial results. [OPEN: repository URL]

Author contributions. [OPEN: CRediT statement]

Tables and figures

Table 1: Sample size per arm required to detect a 2-point difference in RMDQ at 18 weeks with 80% power and two-sided $\alpha = 0.05$, by intra-cluster correlation. Based on a pooled SD of 5.2 points, adjustment for the baseline score (assumed correlation 0.6), 14 analysed participants per practice and a coefficient of variation of practice size of 0.65.

	Intra-cluster correlation			
	0	0.01	0.03	0.05
Design effect (eq. 1)	1.00	1.19	1.56	1.93
Analysed participants per arm	69	82	107	133
Recruited per arm (10% attrition)	76	91	119	148

Table 2: Power to detect a 2-point difference in RMDQ at 18 weeks, two-sided $\alpha = 0.05$, with a t reference distribution on $k_1 + k_2 - 2$ degrees of freedom. Upper panel: the existing 14 practices with 200 analysed participants in total. Lower panel: balanced designs with 14 analysed participants per practice, so that the total sample grows with the number of practices.

		Intra-cluster correlation		
Practices (int./ctrl.)	Analysed N	0.01	0.03	0.05
14 practices, N fixed at 200				
5 / 9	200	0.77	0.65	0.56
6 / 8	200	0.80	0.68	0.59
7 / 7	200	0.80	0.69	0.60
Balanced, 14 participants per practice				
5 / 5	140	0.62	0.51	0.43
6 / 6	168	0.73	0.61	0.52
7 / 7	196	0.80	0.69	0.60
8 / 8	224	0.86	0.76	0.66
9 / 9	252	0.90	0.81	0.72
10 / 10	280	0.93	0.86	0.77

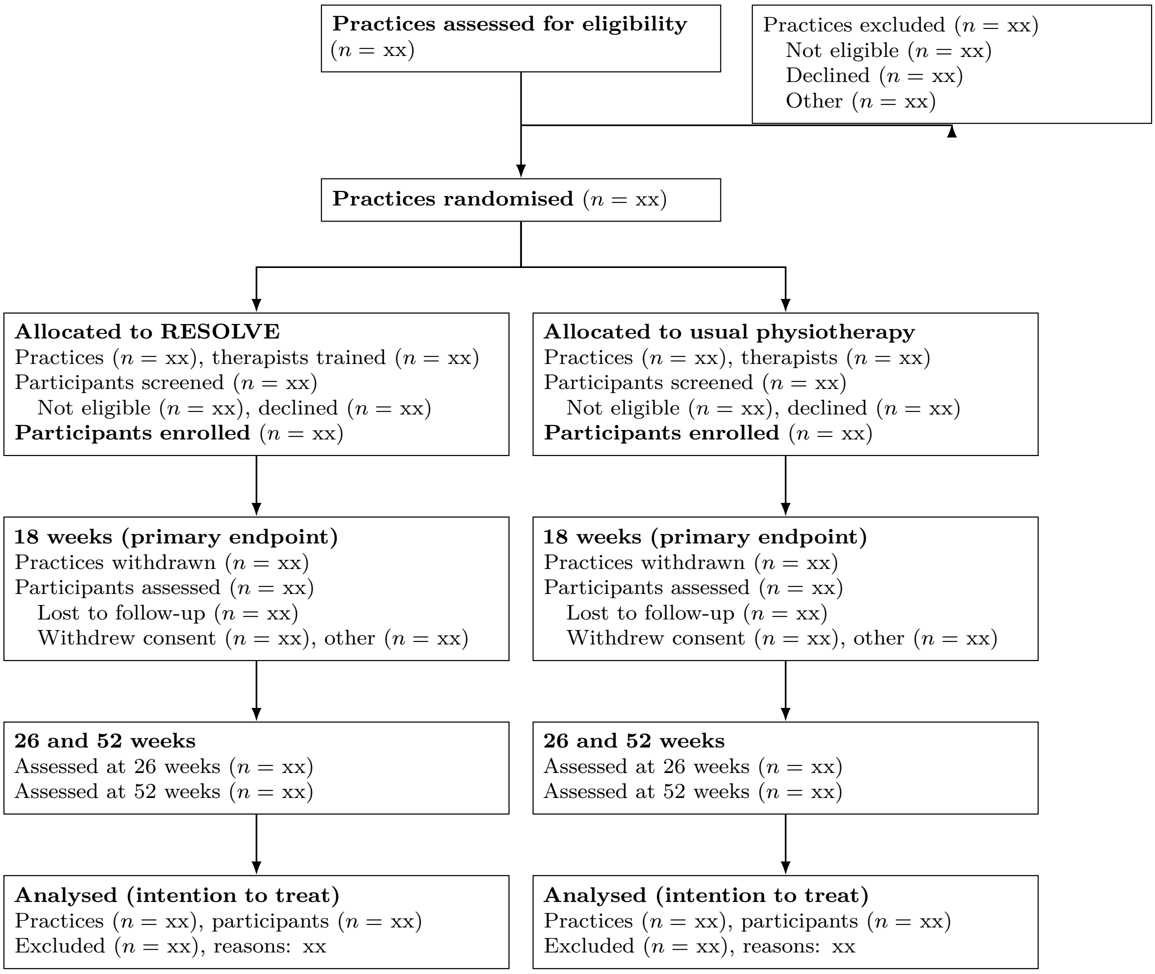

Figure 1: Flow of practices and participants through the trial (shell). Both practices and participants are counted at every stage, as the CONSORT extension for cluster randomised trials requires. The screening counts in the allocation tier are what allow recruitment bias to be judged, since practices know their allocation before they approach patients.

Table 3: Baseline characteristics (shell). Panel A describes the clusters, panel B the participants.

Characteristic	Intervention	Control	All
Panel A: practices			
Number of practices	xx	xx	xx
Therapists per practice, median (IQR)	xx (xx to xx)	xx (xx to xx)	xx (xx to xx)
Participants per practice, median (IQR)	xx (xx to xx)	xx (xx to xx)	xx (xx to xx)
Urban setting, n (%)	xx (xx)	xx (xx)	xx (xx)
Panel B: participants			
Number of participants	xx	xx	xx
Age, mean (SD)	xx (xx)	xx (xx)	xx (xx)
Female, n (%)	xx (xx)	xx (xx)	xx (xx)
Duration of current episode, median (IQR)	xx (xx to xx)	xx (xx to xx)	xx (xx to xx)
RMDQ, mean (SD)	xx (xx)	xx (xx)	xx (xx)
Pain intensity (NRS), mean (SD)	xx (xx)	xx (xx)	xx (xx)
EQ-5D-5L index, mean (SD)	xx (xx)	xx (xx)	xx (xx)
Pain Self-Efficacy Questionnaire, mean (SD)	xx (xx)	xx (xx)	xx (xx)
Tampa Scale of Kinesiophobia, mean (SD)	xx (xx)	xx (xx)	xx (xx)
In paid work, n (%)	xx (xx)	xx (xx)	xx (xx)
Highest education level, n (%)	xx (xx)	xx (xx)	xx (xx)

Table 4: Analysis of the primary outcome (shell). All three analyses estimate the participant-average difference in RMDQ at 18 weeks.

Analysis	Intervention mean (SD)	Control mean (SD)	Difference (95% CI)	*p*
Mixed model, Kenward–Roger (primary)	xx (xx)	xx (xx)	xx (xx to xx)	.xx
Cluster-level, size-weighted	—	—	xx (xx to xx)	.xx
Exact randomisation test	—	—	xx (xx to xx)	.xx
Cluster-average effect (unweighted)	—	—	xx (xx to xx)	.xx
Intra-cluster correlation (95% CI)	xx (xx to xx)			

Table 5: Analysis of secondary outcomes (shell).

Outcome and time point	Intervention mean (SD)	Control mean (SD)	Effect (95% CI)	p
Pain intensity (NRS)				
18 / 26 / 52 weeks	xx (xx)	xx (xx)	xx (xx to xx)	.xx
Patient Specific Functional Scale				
18 / 26 / 52 weeks	xx (xx)	xx (xx)	xx (xx to xx)	.xx
Tactile acuity: two-point discrimination, point-to-point				
18 weeks	xx (xx)	xx (xx)	xx (xx to xx)	.xx
Lumbar movement control				
18 weeks	xx (xx)	xx (xx)	xx (xx to xx)	.xx
Quality of life (EQ-5D-5L)				
18 / 26 / 52 weeks	xx (xx)	xx (xx)	xx (xx to xx)	.xx
Pain knowledge and attitudes (NPQ, Back-PAQ, PSEQ)				
18 / 26 / 52 weeks	xx (xx)	xx (xx)	xx (xx to xx)	.xx
Psychological measures (DASS, TSK)				
18 / 26 / 52 weeks	xx (xx)	xx (xx)	xx (xx to xx)	.xx
Sleep (ISI), body awareness (FreBAQ), goal attainment (GAS)				
18 / 26 / 52 weeks	xx (xx)	xx (xx)	xx (xx to xx)	.xx
Work absence (WPAI), medication intake, adverse events				
18 / 26 / 52 weeks	xx (xx)	xx (xx)	xx (xx to xx)	.xx

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